Answer QUESTION 3 PARTS A and B on this page.

PART A

$$R'(1) \approx \frac{R(2) - R(0)}{2 - 0}$$

$$= \frac{100 - 90}{2 - 0} = \frac{10}{2} = 5$$

$$\boxed{R'(1) \approx 5 \text{ words/min}}$$

PART B

~~$R'(c) = \frac{R(10) - R(0)}{10 - 0} = \frac{162 - 90}{10} = \frac{72}{10}$~~

By the IVT, there must be a time $t = c$ in that, $R(c) = 155$ in the interval $(0, 10)$ because $R(0) = 90 < 155 < 162 = R(10)$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Answer QUESTION 3 PARTS C and D on this page.

PART C

$$\int_0^{10} R(t)\,dt \approx 2\left(\frac{90+100}{2}\right) + 6\left(\frac{100+150}{2}\right) + 2\left(\frac{150+162}{2}\right)$$

$$= 2(95) + 6(125) + 2(156)$$

$$= 190 + 750 + 312$$

$$= 940 + 312$$

$$= 1252 \text{ words}$$

PART D

$$\int_0^{10} -\frac{3}{10}t^2 + 8t + 100\,dt$$

$$= -\frac{1}{10}t^3 + 4t^2 + 100t \Big|_0^{10}$$

$$= -\frac{1}{10}(10)^3 + 4(10^2) + 100(10) - 0$$

$$= -100 + 400 + 1000$$

$$= 1300 \text{ words}$$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.